Diary Data Analysis

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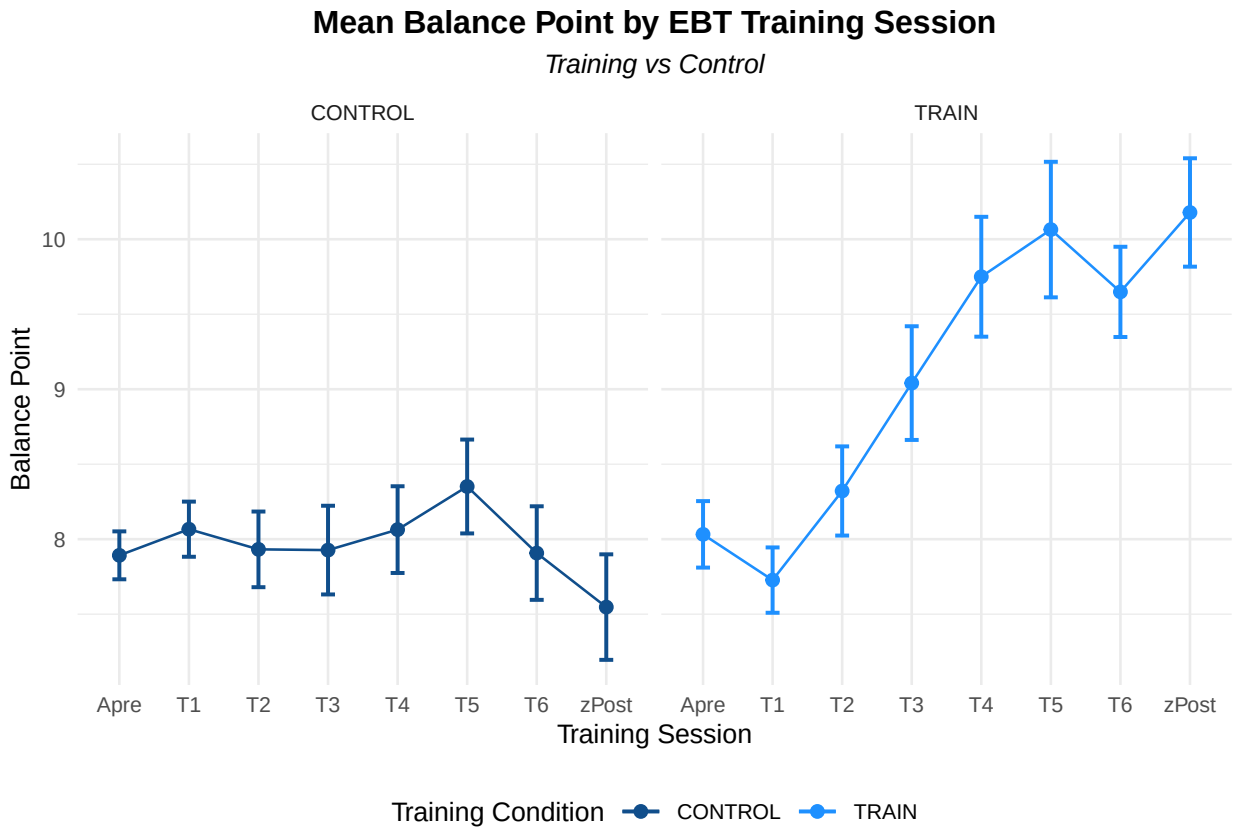
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1) Balance Point

1.1) Plotting data: Impact of training on balance point pre- and post-training

```
## `summarise()` has grouped output by 'Training_Condition'. You can override
## using the `.groups` argument.
```



1.2) 1) Table of mean bp prepost by group

```
## `summarise()` has grouped output by 'aSession'. You can override using the
## `.groups` argument.
```

aSession	Training_Condition	Group_N	Mean_bp	se_bp
Apre	CONTROL	51	7.823529	0.1649858
Apre	TRAIN	52	7.961538	0.2409157
zPost	CONTROL	51	7.607843	0.3587324
zPost	TRAIN	52	9.980769	0.3648285

1.2) Model 1 - Do children differ in their mean balancepoint at baseline?

1.2.1) Linear Regression Model

```
model_1_baseline <- lm(bp ~ Training_Condition, data = nep_baseline_only)
```

1.2.2) Results

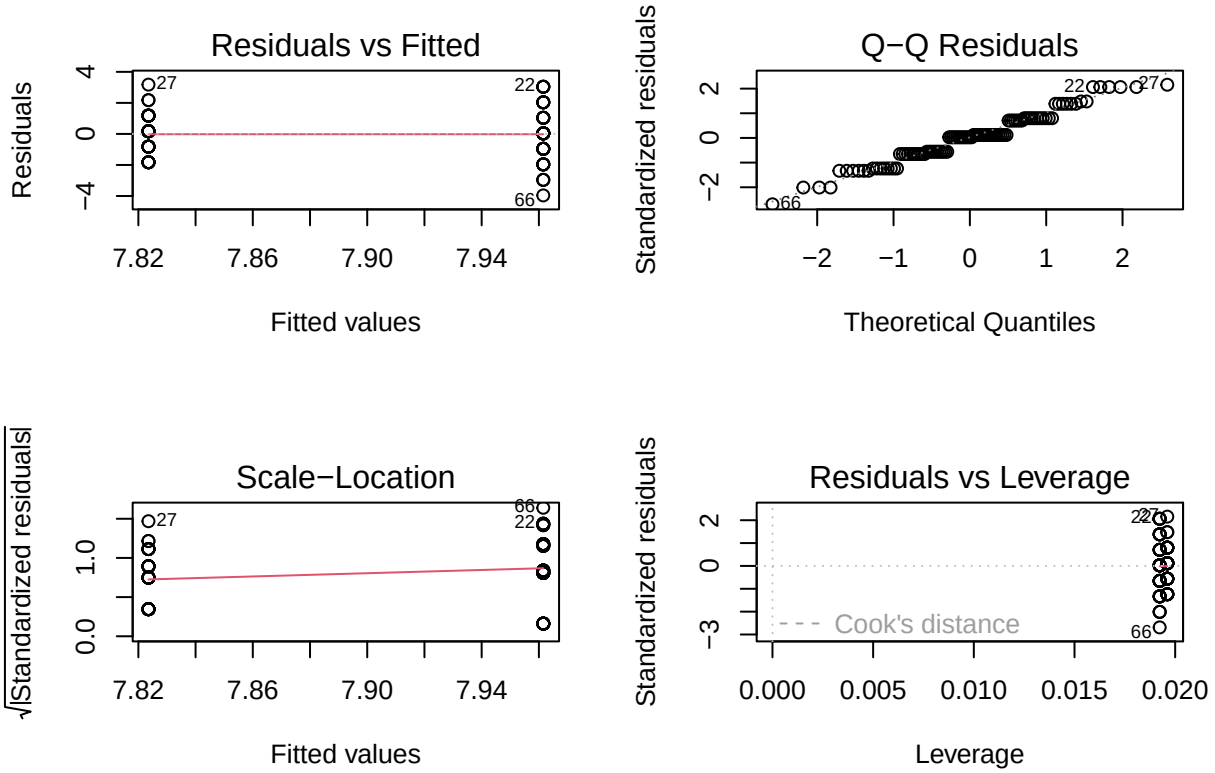
```
##
## Call:
## lm(formula = bp ~ Training_Condition, data = nep_baseline_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9615 -0.9615  0.0385  1.0385  3.1765
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                7.8235     0.2082  37.573   <2e-16 ***
## Training_ConditionTRAIN    0.1380     0.2931   0.471    0.639
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.487 on 101 degrees of freedom
## Multiple R-squared:  0.002191,   Adjusted R-squared:  -0.007688
## F-statistic: 0.2218 on 1 and 101 DF,  p-value: 0.6387
```

1.2.3) 95% Confidence Intervals

```
##                               2.5 %    97.5 %
## (Intercept)              7.4104674 8.2365914
## Training_ConditionTRAIN -0.4433336 0.7193517
```

```
##                              0.25 %  99.75 %
## (Intercept)              7.2259224 8.421136
## Training_ConditionTRAIN -0.7030619 0.979080
```

1.2.4) Assumption Checks:



1.3) Model 2 - Do children differ in their mean balancepoint at at follow up?

Preregistered: Model predicting difference in T2 and T1 BP by IBT

1.3.1) Linear Regression Model

```
lm_bp_diff <- lm(difference_s2_minus_s1 ~ Training_Condition, data = nep_wide_pre_post_wider)
```

1.3.2) Results

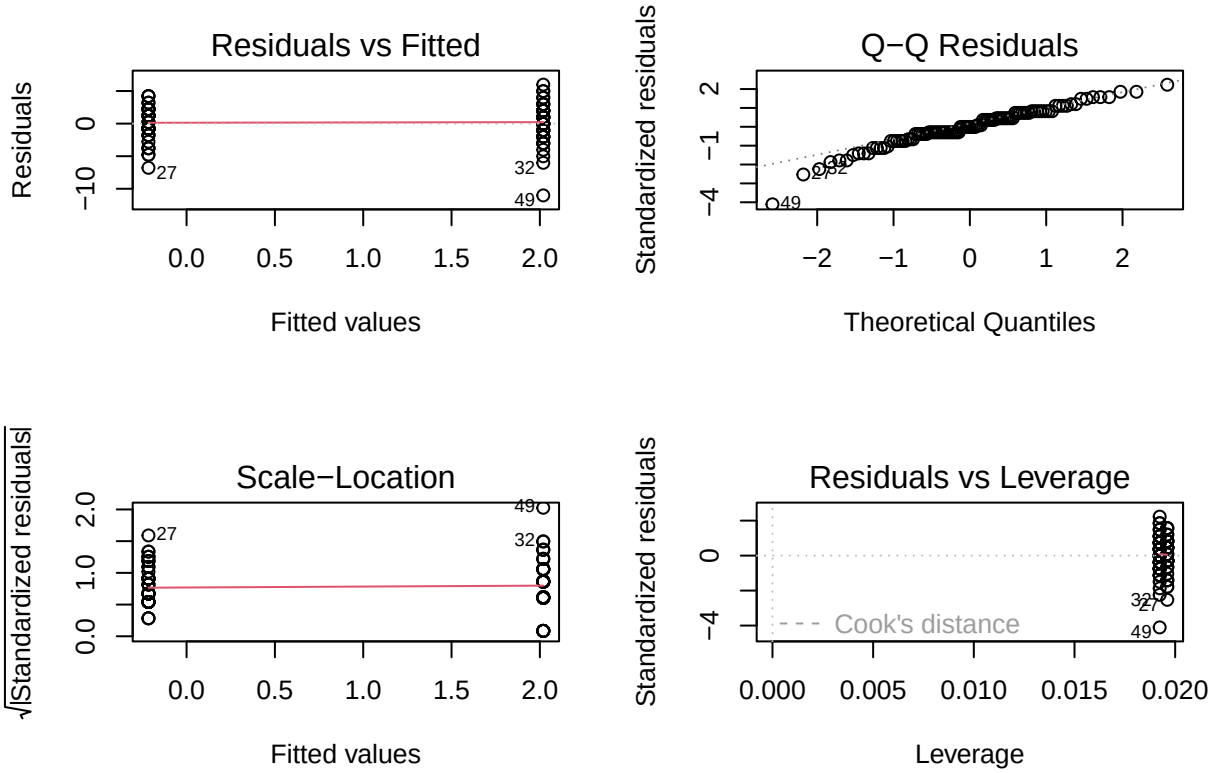
```
##
## Call:
## lm(formula = difference_s2_minus_s1 ~ Training_Condition, data = nep_wide_pre_post_wider)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.0192  -1.0192  -0.0192   1.9808   5.9808
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -0.2157     0.3794  -0.568    0.571
## Training_ConditionTRAIN    2.2349     0.5340   4.185 6.09e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.71 on 101 degrees of freedom
## Multiple R-squared:  0.1478, Adjusted R-squared:  0.1393
## F-statistic: 17.51 on 1 and 101 DF,  p-value: 6.095e-05
```

1.3.3) 95% Confidence Intervals

```
##                              2.5 %    97.5 %
## (Intercept)             -0.9683904 0.5370178
## Training_ConditionTRAIN  1.1755628 3.2942713
```

```
##                             0.25 %   99.75 %
## (Intercept)             -1.3046783 0.8733057
## Training_ConditionTRAIN  0.7022718 3.7675623
```

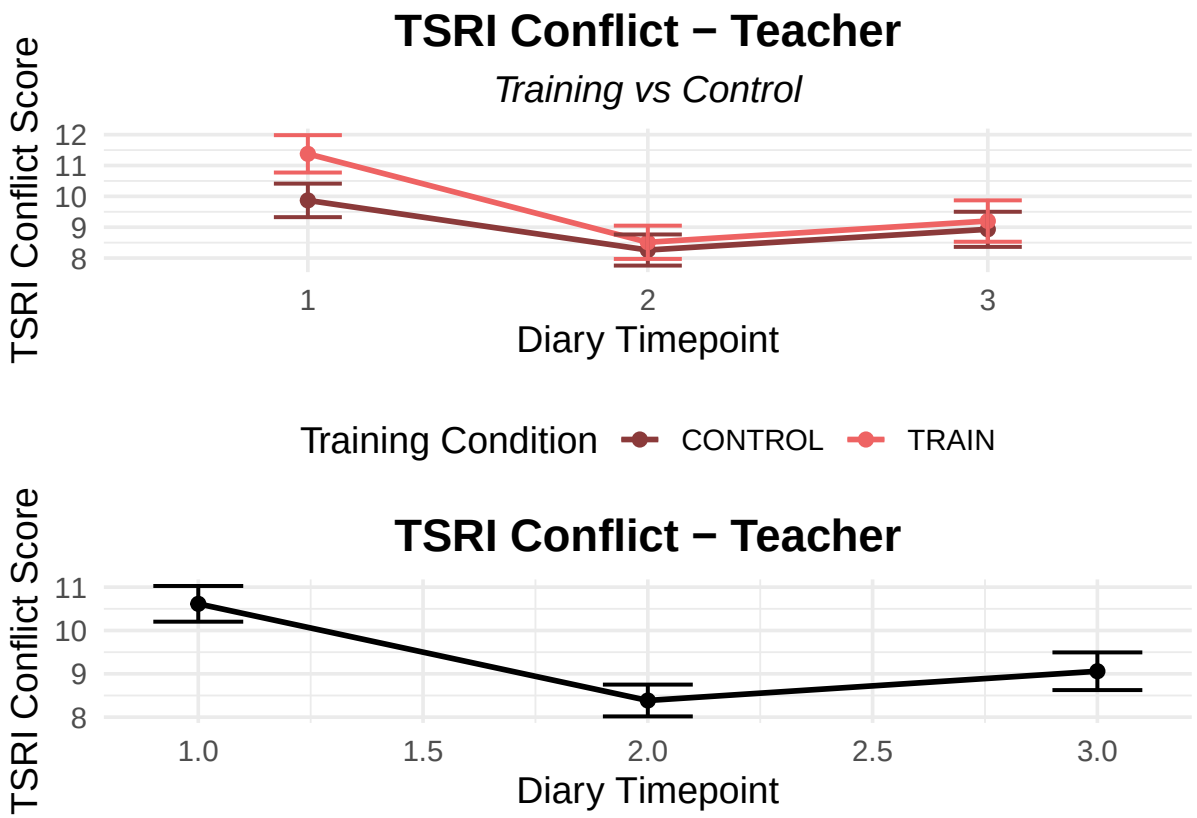
1.3.4) Assumption tests



2) TSRI Conflict

2.1) Summarising data: Do teacher ratings of Conflict at T2 differ by IBT condition?

2.1.1) Plot



2.1.2) Table

Training_Condition	n	M_diff_con_s2_s1	SE_diff_con_s2_s1
CONTROL	46	-1.608696	0.5187110
TRAIN	45	-2.866667	0.6192208

2.2) Model - Does IBT predict Conflict Difference Scores?

2.2.1) Linear Regression

```
model_conflict_t2 <- lm(diff_con_s2_s1 ~ Training_Condition, data = TSRI_con_wide)
```

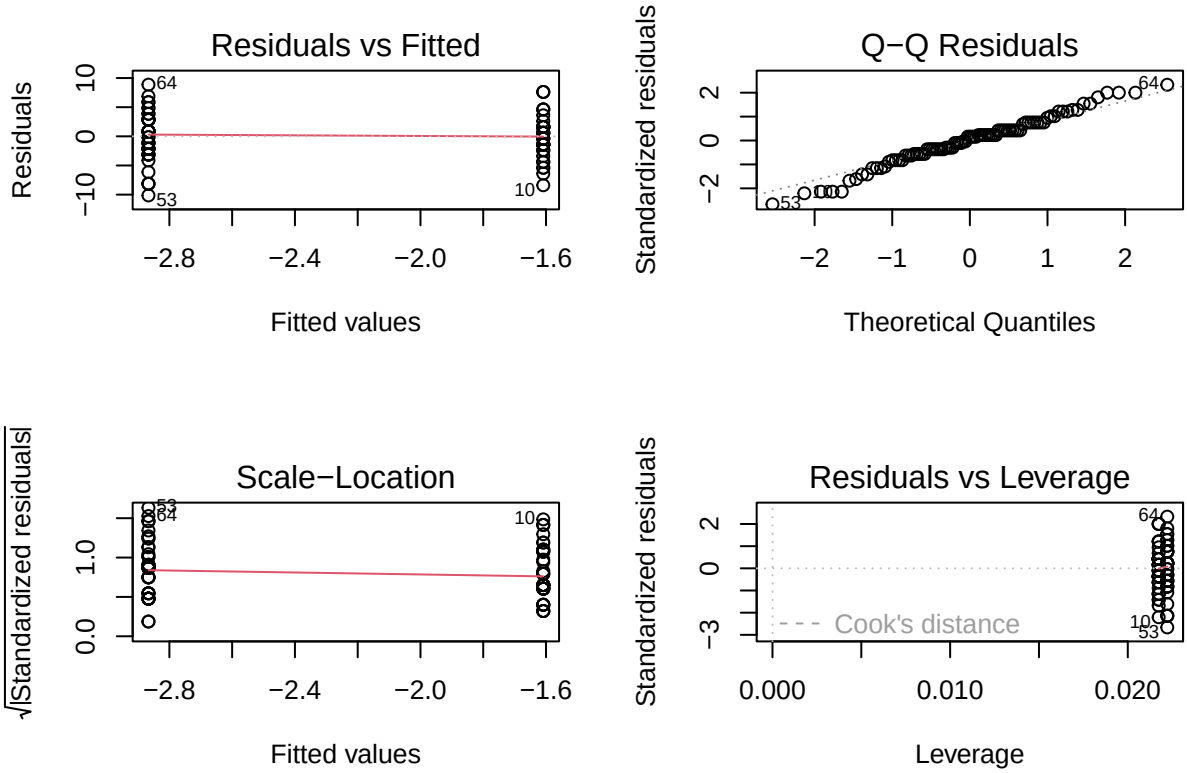
2.2.2) Results

```
##
## Call:
## lm(formula = diff_con_s2_s1 ~ Training_Condition, data = TSRI_con_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1333  -2.1333   0.6087   2.1087   8.8667
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -1.6087     0.5670  -2.837  0.00563 **
## Training_ConditionTRAIN   -1.2580     0.8063  -1.560  0.12226
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.846 on 89 degrees of freedom
## Multiple R-squared:  0.02662,    Adjusted R-squared:  0.01569
## F-statistic: 2.434 on 1 and 89 DF,  p-value: 0.1223
```

2.2.3) Confidence Intervals

```
##                              2.5 %     97.5 %
## (Intercept)              -2.735304 -0.4820871
## Training_ConditionTRAIN -2.860063  0.3441212
```

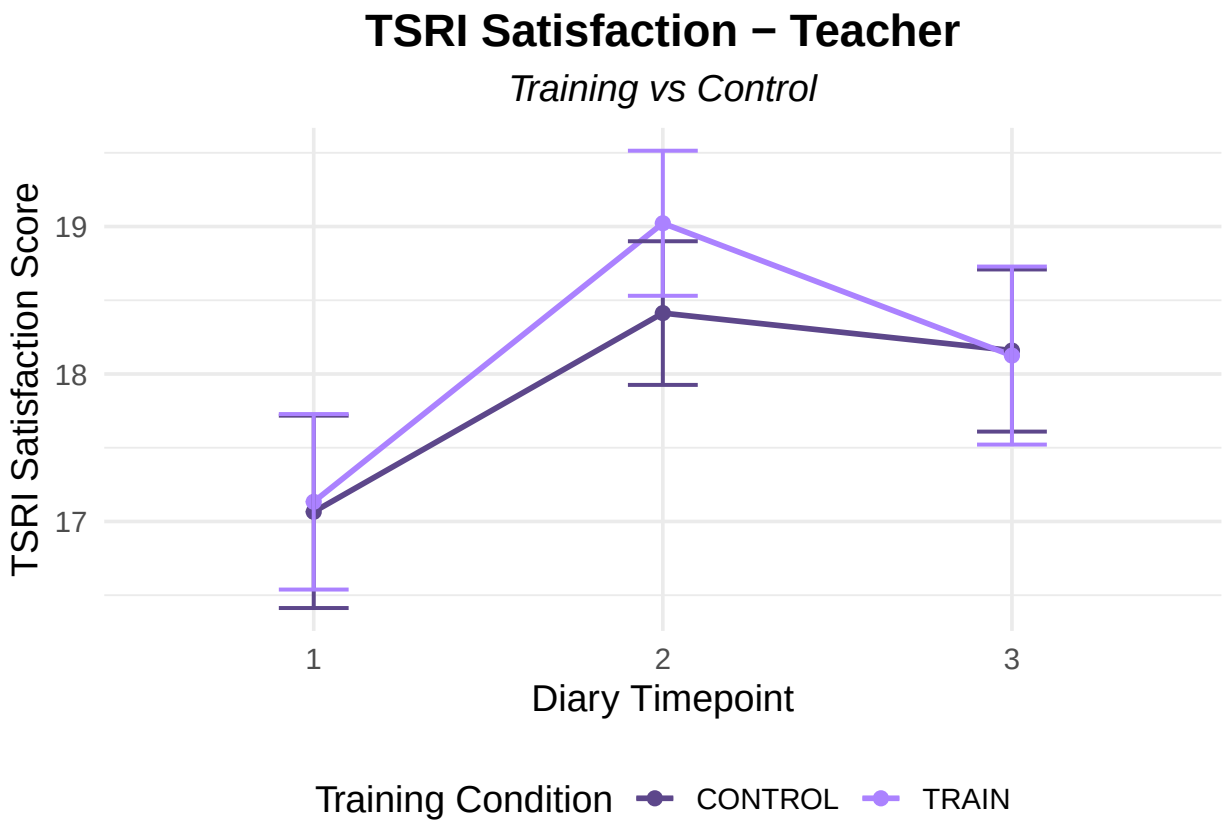
2.2.4) Assumption Check

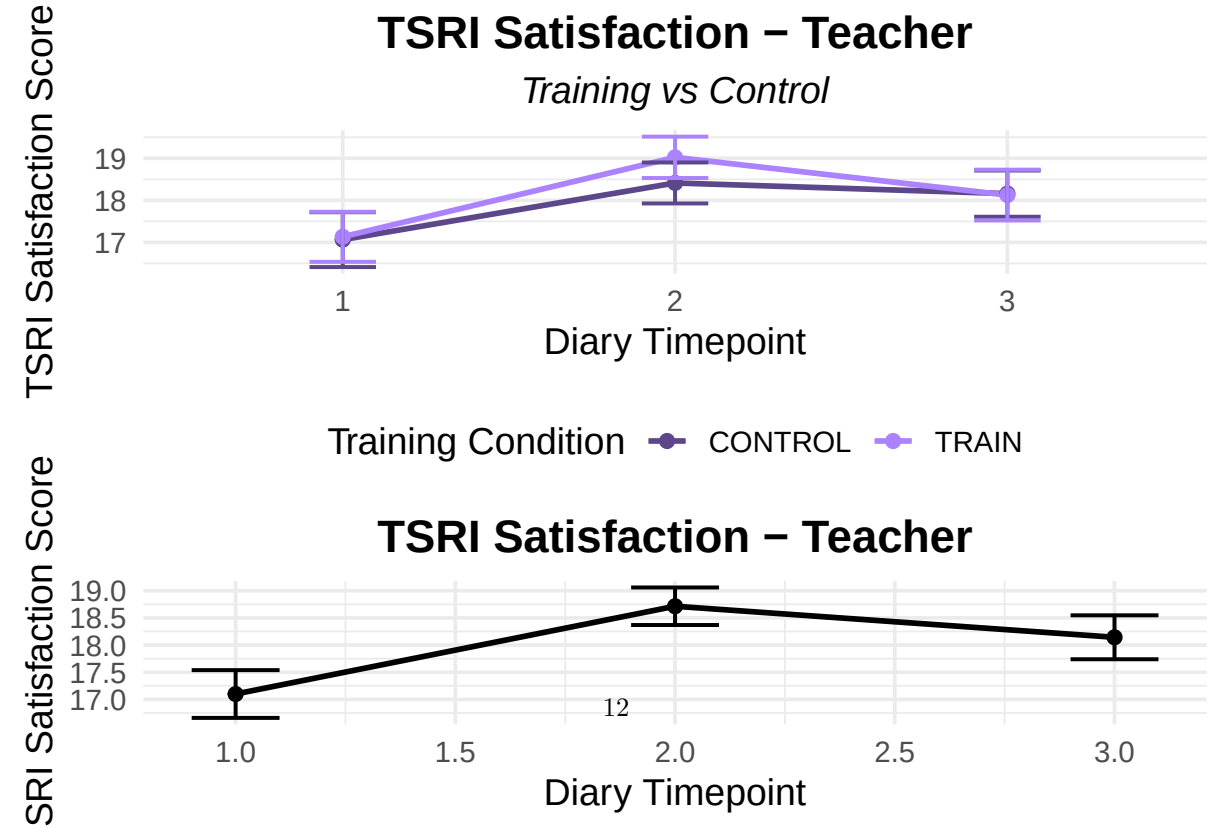


3) TSRI: Satisfaction

3.1) Summariseing Satisfaction Data

3.1.1) Plot: Do teacher ratings of Satisfaction T2 differ by IBT condition?





3.1.2) Table

Training_Condition	n	M_diff_Satisfaction_s2_s1	SE_diff_Satisfaction_s2_s1
CONTROL	46	1.347826	0.5403679
TRAIN	45	1.888889	0.4394926

3.2) Model - Do teacher ratings of Satisfaction T2 differ by IBT condition?

3.2.1) Linear Regression

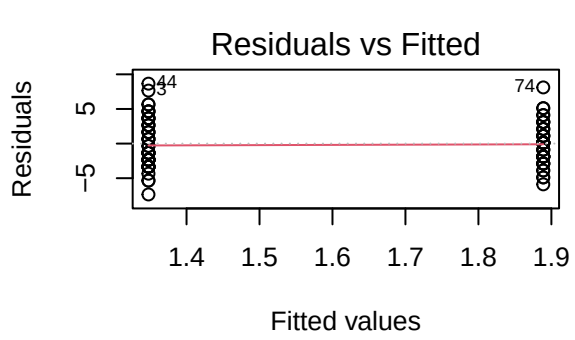
```
model_satisfaction_t2 <- lm(diff_Satisfaction_s2_s1 ~ Training_Condition, data = TSRI_satisfaction_wide)
```

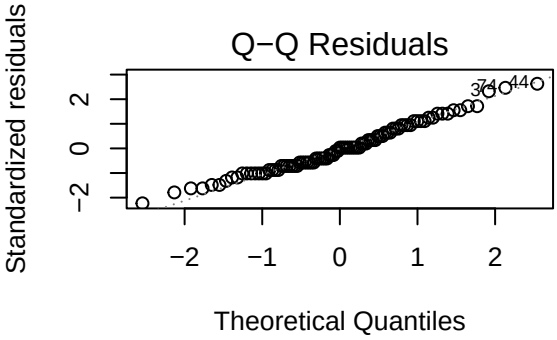
3.2.2) Results

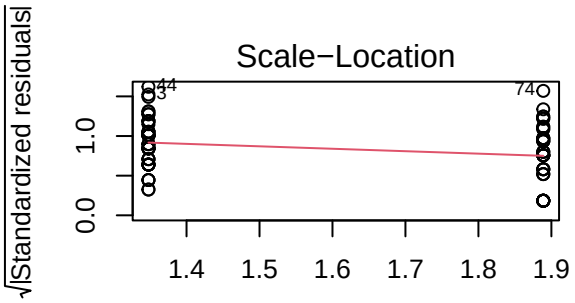
```
##
## Call:
## lm(formula = diff_Satisfaction_s2_s1 ~ Training_Condition, data = TSRI_satisfaction_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.3478 -2.3478  0.1111  2.3816  8.6522
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                1.3478     0.4910   2.745  0.00732 **
## Training_ConditionTRAIN    0.5411     0.6982   0.775  0.44042
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.33 on 89 degrees of freedom
## Multiple R-squared:  0.006703,   Adjusted R-squared:  -0.004458
## F-statistic: 0.6006 on 1 and 89 DF,  p-value: 0.4404
```

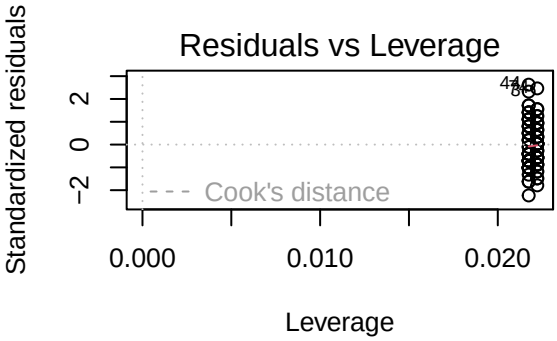
3.2.3) 95% Confidence Itnervalts

```
##                               2.5 %   97.5 %
## (Intercept)              0.3722721 2.323380
## Training_ConditionTRAIN -0.8462223 1.928348
```


Residuals vs Fitted
Residuals
Fitted values
44
3
74
−5
5
1.4
1.5
1.6
1.7
1.8
1.9


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
74
44
−2
0
2
−2
−1
0
1
2

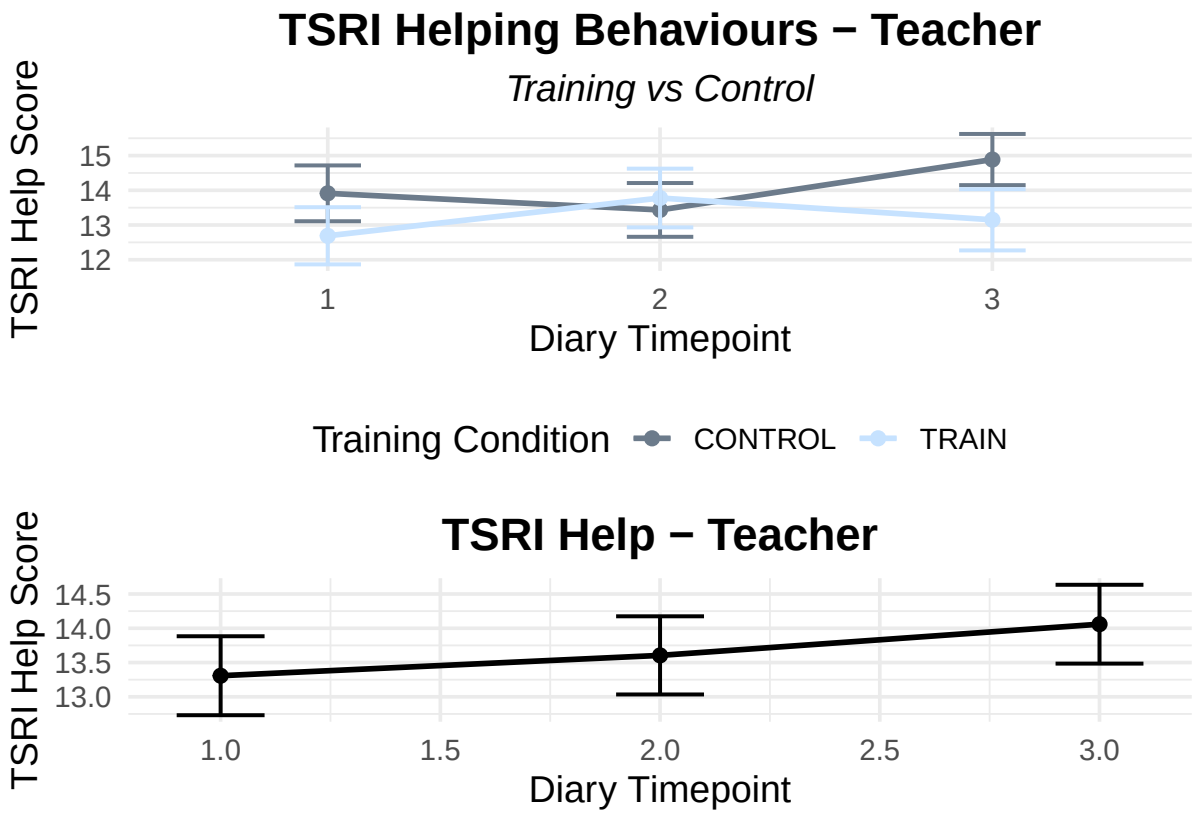

Scale−Location
√|Standardized residuals|
Fitted values
44
3
74
0.0
1.0
1.4
1.5
1.6
1.7
1.8
1.9


Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
−2
0
2
0.000
0.010
0.020

4) TSRI: Helping behaviours

4.1) Summarising data: Do teacher ratings of Help at T2 differ by IBT condition?

4.1.1) Plot:



4.1.2) Table

Training_Condition	n	M_diff_Help_s2_s1	SE_diff_Help_s2_s1
CONTROL	46	-0.4782609	0.4762364
TRAIN	45	1.0888889	0.5841529

4.2) Model: Do teacher ratings of Help at T2 differ by IBT condition?

4.2.1) Linear Regression

```
model_help_t2 <- lm(diff_Help_s2_s1 ~ Training_Condition, data = TSRI_help_wide)
```

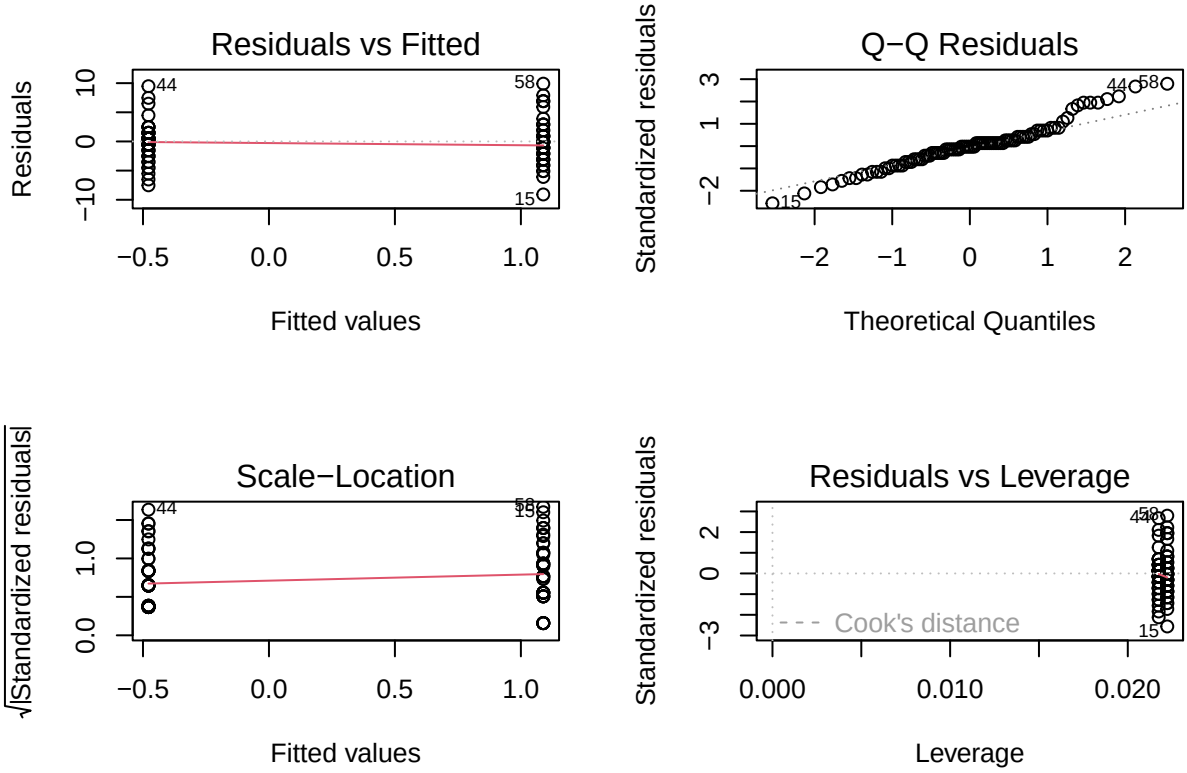
4.2.2) Results

```
##
## Call:
## lm(formula = diff_Help_s2_s1 ~ Training_Condition, data = TSRI_help_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.0889 -2.0889 -0.0889  1.4783  9.9111
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -0.4783     0.5289  -0.904   0.3683
## Training_ConditionTRAIN    1.5671     0.7521   2.084   0.0401 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.587 on 89 degrees of freedom
## Multiple R-squared:  0.04652,    Adjusted R-squared:  0.0358
## F-statistic: 4.342 on 1 and 89 DF,  p-value: 0.04005
```

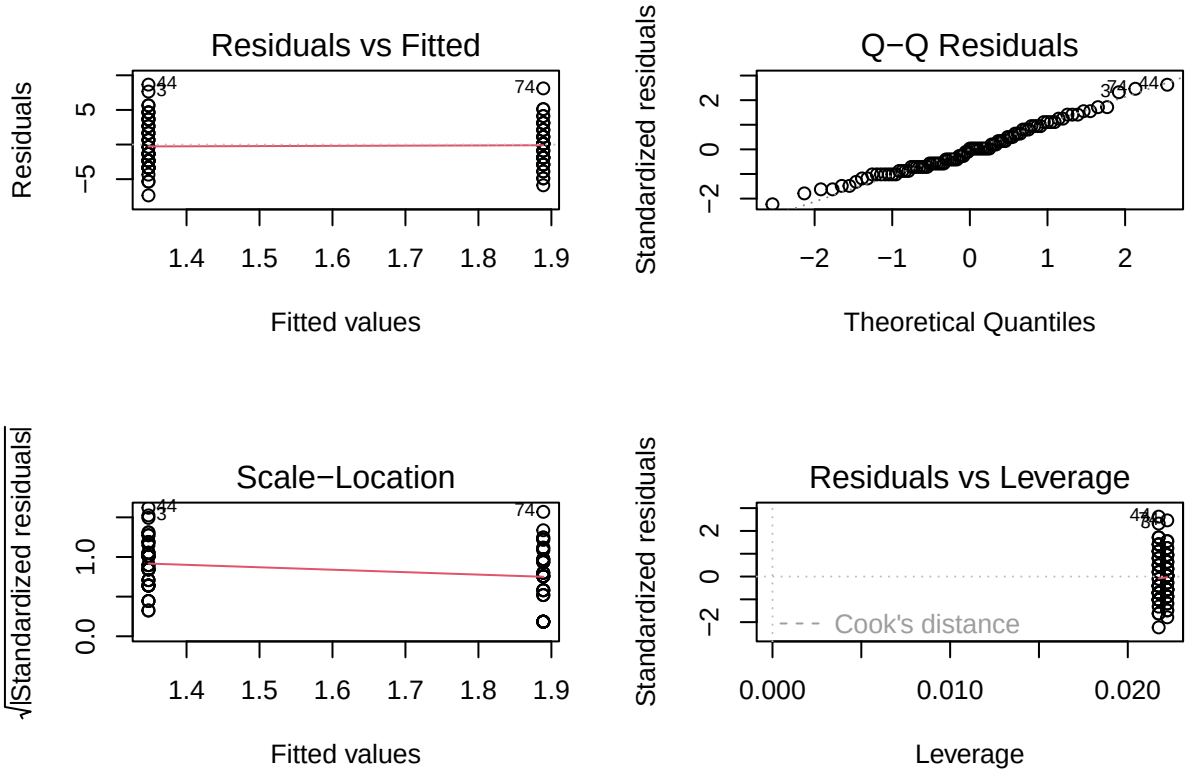
4.2.3) Confidence Intervals

```
##                               2.5 %    97.5 %
## (Intercept)             -1.52912194 0.5726002
## Training_ConditionTRAIN  0.07277424 3.0615253
```

```
##                              0.25 %  99.75 %
## (Intercept)             -2.0007295 1.044208
## Training_ConditionTRAIN -0.5978746 3.732174
```


Residuals vs Fitted
Residuals
Fitted values
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage

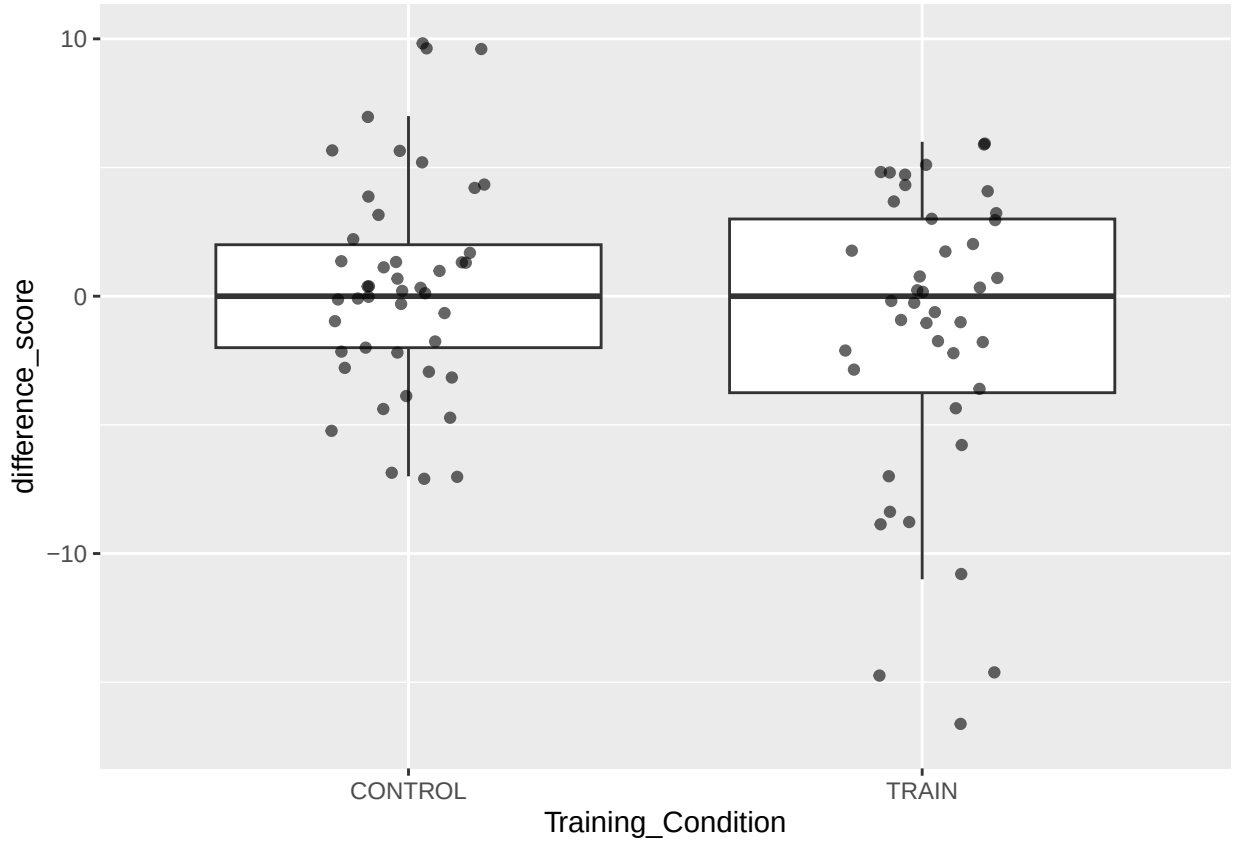
4.2.4) Assumption checks



5) Child SRQ-A Prosocial Intentions.

5.1) Sumarising child SRQ data

5.1.1) Plotting child SRQ difference scores



5.2.2) Table

Training_Condition	n	M_diff_difference_score	SE_diff_difference_score
CONTROL	45	0.4888889	0.6281464
TRAIN	42	-1.4047619	0.9049224

5.2) Model: child SRQ Difference Scores

5.2.1) Linear Regression

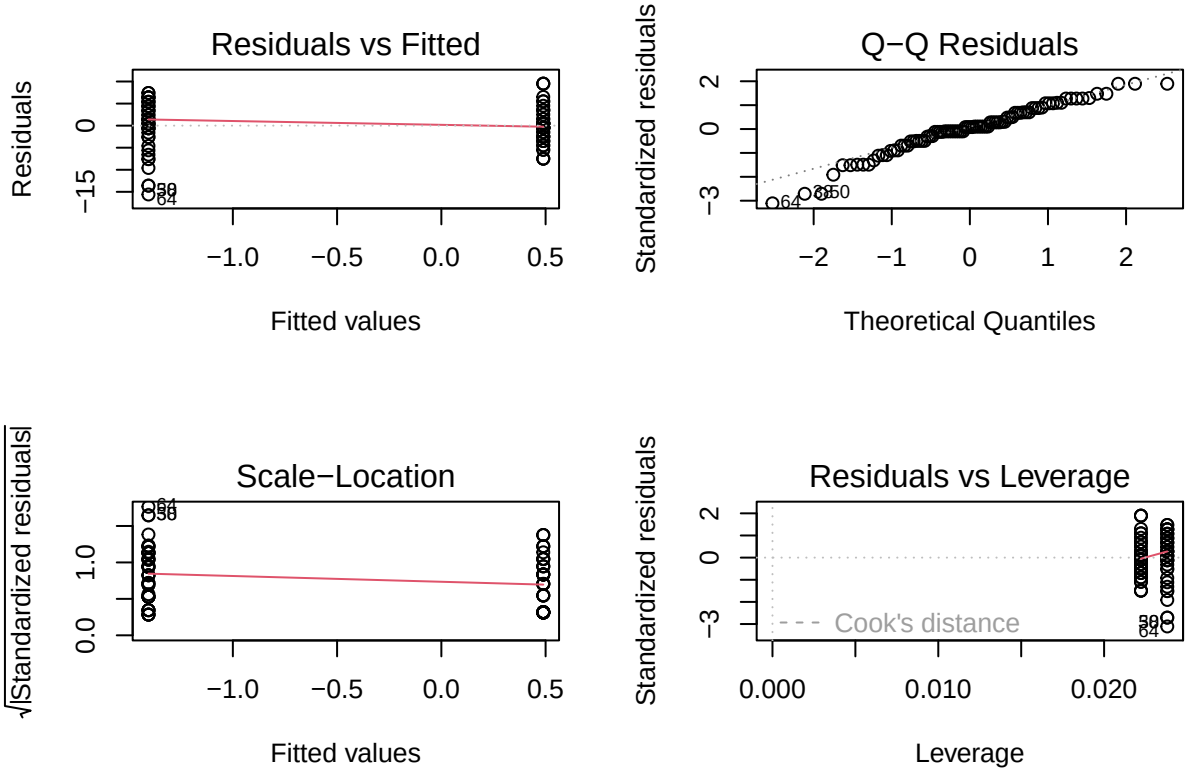
```
# S2 - S1 so we want the score to be positive
model_srq_diff <- lm(difference_score ~ Training_Condition, data = child_srq)
```

5.2.2) Results

```
##
## Call:
## lm(formula = difference_score ~ Training_Condition, data = child_srq)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.5952  -2.4889   0.4048   3.4579   9.5111
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                0.4889     0.7569   0.646   0.5201
## Training_ConditionTRAIN   -1.8937     1.0894  -1.738   0.0858 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.077 on 85 degrees of freedom
## Multiple R-squared:  0.03433,    Adjusted R-squared:  0.02297
## F-statistic: 3.022 on 1 and 85 DF,  p-value: 0.08578
```

5.2.3) 95% Confidence Intervals

```
##                              2.5 %    97.5 %
## (Intercept)             -1.016041 1.9938192
## Training_ConditionTRAIN -4.059615 0.2723139
```


Residuals vs Fitted
Residuals
Fitted values
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage

6) Exploratory analyses

6.1) (exploratory) Do teacher ratings of conflict differ from T1-3

Number of participants:

```
length(unique(TSRI_ICU_d1d2d3_full$ID_numeric))
```

```
## [1] 84
```

```
model_conflict_exploratory_simpler <- lm(T_TSRI_Conflict_Total ~ diaryTP , data = TSRI_ICU_d1d2d3_full)


summary(model_conflict_exploratory_simpler)
```

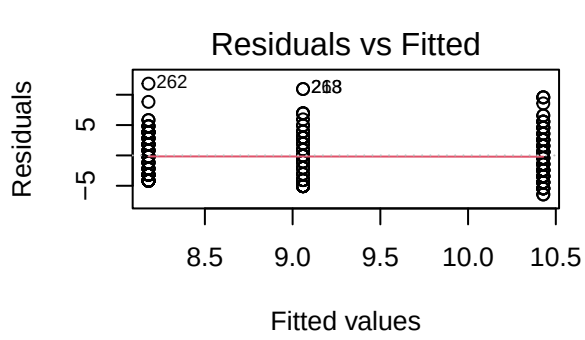
```
##
## Call:
## lm(formula = T_TSRI_Conflict_Total ~ diaryTP, data = TSRI_ICU_d1d2d3_full)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.4286 -3.1786 -0.1786  2.9405 11.8214
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  10.4286     0.4179  24.952  < 2e-16 ***
## diaryTP2     -2.2500     0.5911  -3.807 0.000177 ***
## diaryTP3     -1.3690     0.5911  -2.316 0.021358 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.831 on 249 degrees of freedom
## Multiple R-squared:  0.05581,    Adjusted R-squared:  0.04823
## F-statistic: 7.359 on 2 and 249 DF,  p-value: 0.000785
```

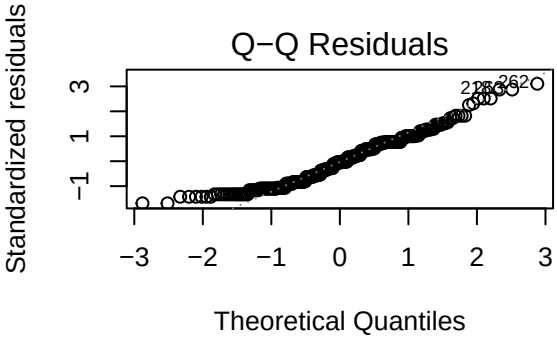
```
confint(model_conflict_exploratory_simpler)
```

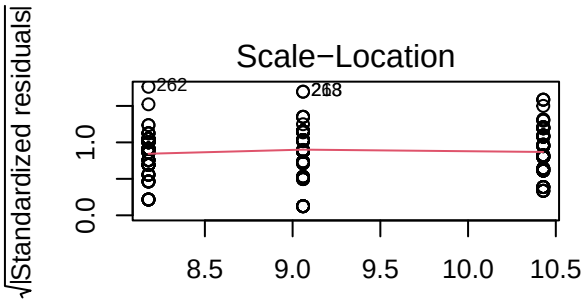
```
##                 2.5 %     97.5 %
## (Intercept)  9.605410 11.2517329
## diaryTP2    -3.414126 -1.0858739
## diaryTP3    -2.533174 -0.2049216
```

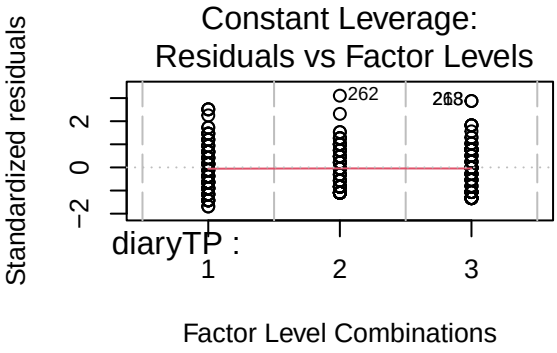
```
confint(model_conflict_exploratory_simpler, level = 0.995)
```

```
##                0.25 %    99.75 %
## (Intercept)  9.244835 11.6123082
## diaryTP2    -3.924057 -0.5759434
## diaryTP3    -3.043104  0.3050090
```


Residuals vs Fitted
Residuals
Fitted values
262
8.5
9.0
9.5
10.0
10.5
−5
5


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
262
−3
−2
−1
0
1
2
3


Scale−Location
√|Standardized residuals|
Fitted values
262
8.5
9.0
9.5
10.0
10.5
0.0
1.0

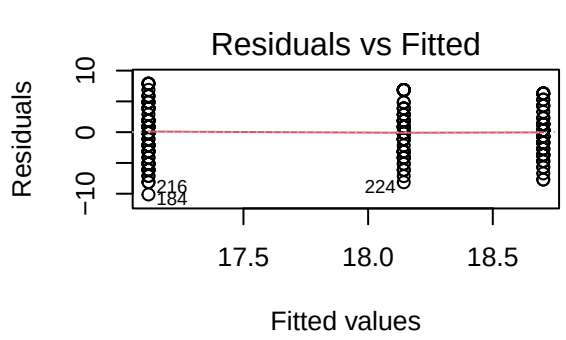

Constant Leverage:
Residuals vs Factor Levels
Standardized residuals
Factor Level Combinations
diaryTP :
262
1
2
3
−2
0
2

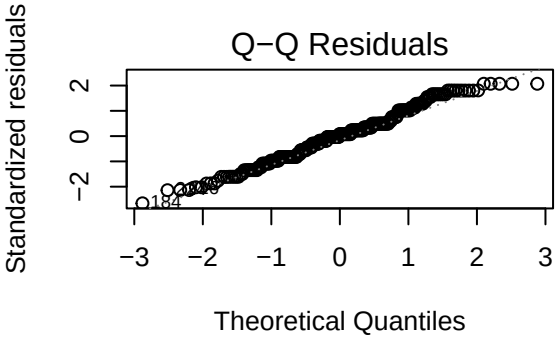
6.2) (exploratory) Do teacher ratings of satisfaction differ from T1-3

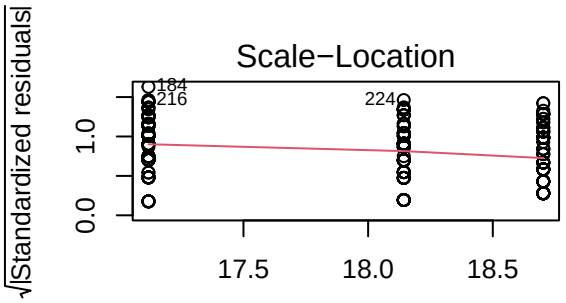
```
##
## Call:
## lm(formula = T_TSRI_Satisfaction_Total ~ diaryTP, data = TSRI_ICU_d1d2d3_full)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1190  -3.1190   0.2976   1.8810   7.8810
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  17.1190     0.4179  40.965  < 2e-16 ***
## diaryTP2      1.5833     0.5910   2.679  0.00787 **
## diaryTP3      1.0238     0.5910   1.732  0.08445 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.83 on 249 degrees of freedom
## Multiple R-squared:  0.0288, Adjusted R-squared:  0.021
## F-statistic: 3.692 on 2 and 249 DF,  p-value: 0.0263
```

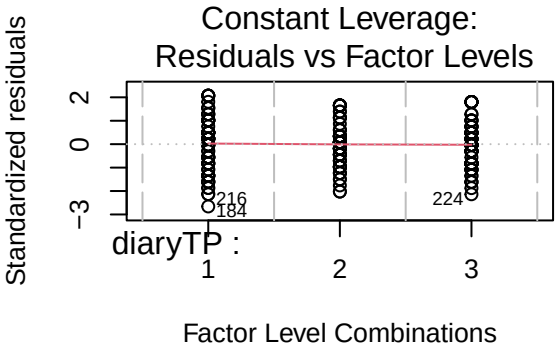
```
##                  2.5 %    97.5 %
## (Intercept) 16.2959921 17.942103
## diaryTP2     0.4193571  2.747310
## diaryTP3    -0.1401667  2.187786
```

```
##                  0.25 %   99.75 %
## (Intercept) 15.93546320 18.302632
## diaryTP2    -0.09050781  3.257174
## diaryTP3    -0.65003162  2.697651
```


Residuals vs Fitted
Residuals
Fitted values
216
184
224


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
184


Scale−Location
√|Standardized residuals|
Fitted values
184
216
224

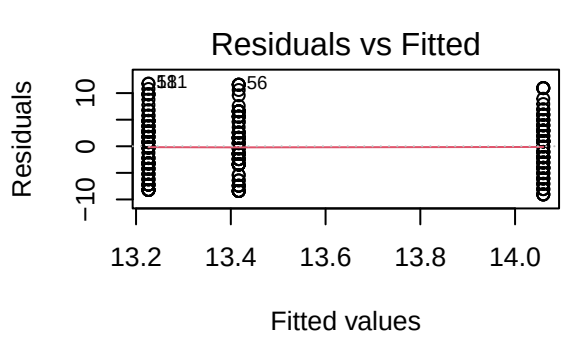

Constant Leverage:
Residuals vs Factor Levels
Standardized residuals
diaryTP :
Factor Level Combinations
216
184
224

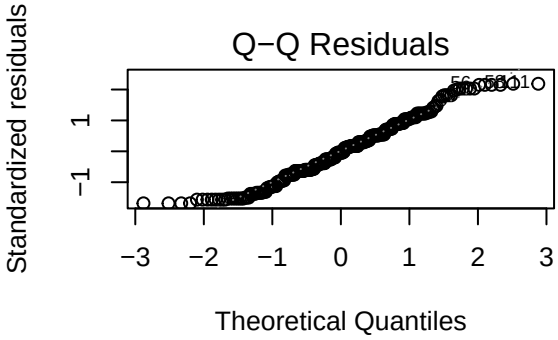
6.3) (exploratory) Do teacher ratings of help differ from T1-3

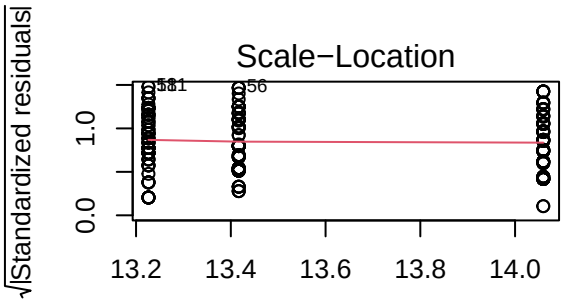
```
##
## Call:
## lm(formula = T_TSRI_Help_Total ~ diaryTP, data = TSRI_ICU_d1d2d3_full)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.0595 -3.5774 -0.2262  3.7738 11.7738
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  13.2262     0.5911  22.376   <2e-16 ***
## diaryTP2      0.1905     0.8359   0.228     0.82
## diaryTP3      0.8333     0.8359   0.997     0.32
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.417 on 249 degrees of freedom
## Multiple R-squared:  0.004364,   Adjusted R-squared:  -0.003633
## F-statistic: 0.5457 on 2 and 249 DF,  p-value: 0.5801
```

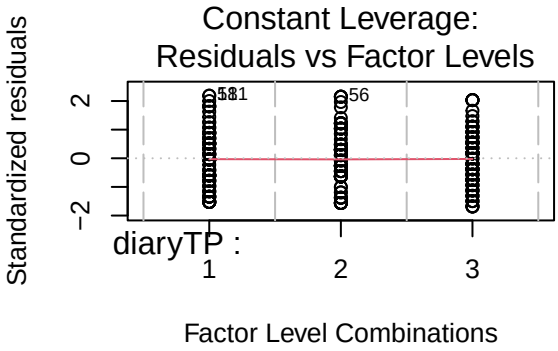
```
##                  2.5 %    97.5 %
## (Intercept) 12.0620142 14.390367
## diaryTP2    -1.4559178  1.836870
## diaryTP3    -0.8130606  2.479727
```

```
##             -4900 % 5000 %
## (Intercept)     NaN    NaN
## diaryTP2        NaN    NaN
## diaryTP3        NaN    NaN
```


Residuals vs Fitted
Residuals
Fitted values
13.2
13.4
13.6
13.8
14.0
−10
0
10
56


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
−3
−2
−1
0
1
2
3
56


Scale−Location
√|Standardized residuals|
Fitted values
13.2
13.4
13.6
13.8
14.0
0.0
1.0
56


Constant Leverage:
Residuals vs Factor Levels
Standardized residuals
diaryTP :
Factor Level Combinations
1
2
3
−2
0
2
56